

Total Score

Name

School

Roll No

Somalilandschools.Online

FREE EXAM PAPERS,
LESSONS, NOTES AND

TEXT BOOKS

Contact:

00 252 63 6551739

00 252 63 4421900

00 252 65 4421900

Republic of Somaliland

Somaliland National Examination Board

Form Four

PHYSICS EXAMINATION

July 2017

TIME 2 HOURS

Plus 10 minutes for reading through the paper

INSTRUCTIONS TO CANDIDATES

This paper consists of 13 printed pages.

Count them now. Inform the invigilator if there are any pages missing.

PART 1: 20 Multiple Choice Questions 20 Marks

PART 2: 8 Structured Questions 80 Marks

TOTAL 100 Marks

- Answer ALL questions in Part 1 and 2.
- No extra papers are allowed.

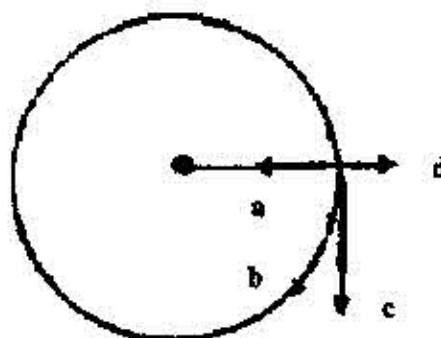
Blank page.

PART 1: ANSWER ALL QUESTIONS IN THIS SECTION.
CIRCLE THE CORRECT ANSWER.

(20 Marks)

1. Velocity is defined as :
 - A. Distance moved in unit time
 - B. Displacement in unit time
 - C. Acceleration in particular direction
 - D. Energy transferred per unit time.
2. The quantity that tells how hard a force acts is known as :
 - A. Momentum
 - B. Impulse
 - C. Moment
 - D. Torque
3. In launching a rocket , hot gases of high pressure is ejected downwards and as a result the rocket moves upwards. This is an example of Newton's _____ law of motion.
 - A. First
 - B. Second
 - C. Third
 - D. Fourth.
4. You are sitting in a car.It stops suddenly and you fall forwards. This is an example of your:
 - A. Momentum
 - B. Inertia
 - C. Impulse
 - D. Torque
5. When the forces acting on a moving body are balanced the net force becomes zero. As a result the body will :
 - A. Move with an increasing velocity
 - B. Move with decreasing velocity
 - C. Move with constant (Uniform) velocity
 - D. Stop moving

6. A bullet is fired with an initial velocity of 50 m/s at an angle of 30° to the horizontal. What is the horizontal component of bullet's initial velocity?
- 15 m/s
 - 25 m/s
 - 44.3 m/s
 - 50 m/s
7. Ahmed whirls a stone along a horizontal circle with a string. If the string breaks, the stone will :
- Move inwards along the radius
 - Continue to move along the curve
 - Move along the tangent
 - Move outwards along the radius
8. SI unit of angular velocity is:
- Newton per radian
 - Radians per second
 - Radians per meter
 - Radians per second squared
9. A girl climbs to the top of high mountain in 15 minutes. If she does 180000 J of energy against gravity, what average power is developed?
- 50 watts
 - 100 watts
 - 120 watts
 - 200 watts
10. $F = -Kx$ is mathematical expression for :
- Newton's second law of motion
 - Hooke's law
 - Snell's law
 - Lenz's law



Somalilandschools.Online

**FREE EXAM PAPERS,
LESSONS, NOTES AND
TEXT BOOKS**

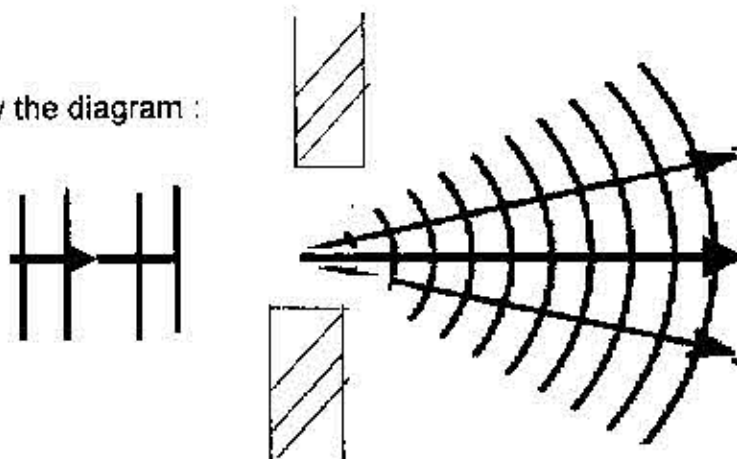
Contact:

00 252 63 6551739

00 252 63 4421900

00 252 65 4421900

11. Sky waves are used to carry information over long distances with the absence of satellites. This is because sky waves :
- A. Can pass through vacuum
 - B. Are reflected by ionospheres
 - C. Can be absorbed by air particles
 - D. Are electromagnetic waves
12. R.F current is produced by an oscillatory circuit. This contains:
- A. Amplifier and capacitor
 - B. Transistor and capacitor
 - C. Capacitor and coil
 - D. Capacitor and diode
13. Process of converting electronic data or information radio systems into a form that people can understand easily is known as:
- A. Encoding
 - B. Decoding
 - C. Translating
 - D. Data organization
14. Which property of waves is illustrated by the diagram :
- A. Reflection
 - B. Refraction
 - C. Diffraction
 - D. Interference



15. A nuclide of mass number A emits an alpha particle . What is the mass number of the resulted nucleus in terms of A :
- A. $A - 2$
 - B. $A - 4$
 - C. $A - 8$
 - D. $A + 1$

16. Which of the following is **NOT** nuclear radiation :

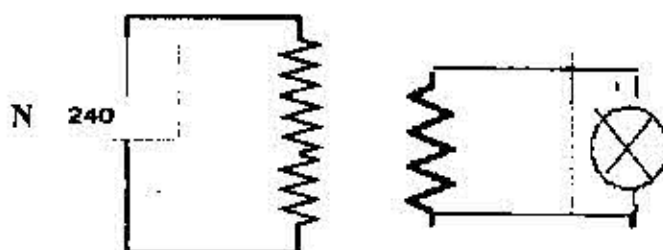
- A. Alpha rays
- B. Beta rays
- C. X-rays
- D. Gamma rays

17. A radioactive sample has half life of 3 months. what fraction of the sample remains undecayed after one year:

- A. $\frac{1}{8}$
- B. $\frac{1}{16}$
- C. $\frac{1}{32}$
- D. $\frac{1}{64}$

18. The figure shows a transformer that steps down mains supply (240V) – what is the voltage across the lamp? The transformer has turns ratio of 30:1

- A. 1 V
- B. 8 V
- C. 80 V
- D. 240 v



19. Which of the following is an example of electric generator :

- A. Loudspeaker
- B. Circuit breaker
- C. Microphone
- D. Galvanometer

20. Electric power transmission is always done using a.c rather than d.c .This is because a.c :
- A. Can be generated easily
 - B. Can be changed easily
 - C. Is safer than d.c
 - D. Is cheaper than d.c

PART TWO. STRUCTURED QUESTIONS. ANSWER ALL QUESTIONS

1. a) Define acceleration and give its unit. (2Marks)

.....

.....

- b) A car increases its velocity from 10 m/s to 30 m/s in 5 s. Calculate?

- i) The acceleration of the car: (3 Marks)

.....

.....

.....

.....

.....

- ii) Distance travelled: (3 Marks)

.....

.....

.....

.....

.....

- iii) Average velocity: (2 Marks)

.....

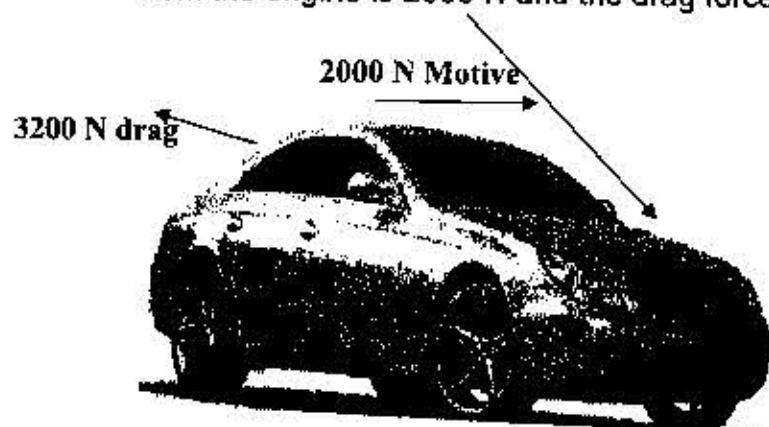
.....

.....

.....

.....

2. The figure shows the forces acting on a car of mass 800 kg. The motive force from the engine is 2000 N and the drag force at this instant is 3200 N.



- a) Explain why the car is decelerating. (2 Marks)

.....

.....

.....

- b) Work out the net force acting on the car ? (2Marks)

.....

.....

.....

- c) Workout the deceleration of the car ? (3 Marks)

.....

.....

.....

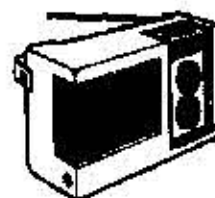
- d) What would happen to the motion of the car if the force from the engine balances the total drag force? (2 Marks)

.....

.....

.....

3. Ahmed is listening to a radio. Complete the passage below using words from the following list :



RF carrier	Resonance	Tuner	Sound	
Aerial	Amplifier	Energy	Electrical	Frequency

- A. The of radio receiver detect signals from many different radio stations and converts into electrical signals.
- B. TheSelects (picks up) one particular frequency from a given station by
- C. The demodulator decodes the incoming signals by removing theleaving only AF signals.
- D. The increases the amplitude of these electrical signals.
- E. Therequired to do this is supplied by the battery.
- F. The loudspeaker convertsenergy intoenergy.

(8x1.5 = 12 Marks)

Somalilandschools.Online

**FREE EXAM PAPERS,
LESSONS, NOTES AND**

TEXT BOOKS

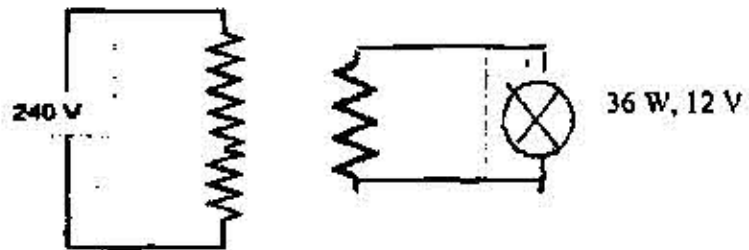
Contact:

00 252 63 6551739

00 252 63 4421900

00 252 65 4421900

4. The figure shows a transformer to step down mains supply to light up the 36 w, 12 v the lamp.



- A. What is meant by step down transform ? (2 Marks)

.....
.....

- B. What is the turn's ratio of the transformer windings? (2Marks)

.....
.....

- C. What current flows through the lamp? (3 Marks)

.....
.....
.....

- D. What current flows through the primary coil? Assume no power loss in the transformer? (3 marks)

.....
.....
.....

Somalilandschools.Online

**FREE EXAM PAPERS,
LESSONS, NOTES AND**

TEXT BOOKS

Contact:

00 252 63 6551739

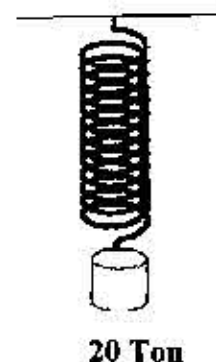
00 252 63 4421900

00 252 65 4421900

5. Match the terms in column A with their descriptions in column B. (10x1 = 10 Marks)

	A		B
1	Transverse _____	A	Maximum displacement from undisturbed position
2	Longitudinal _____	B	The highest point of oscillating particles
3	Crest _____	C	Particle oscillation is perpendicular to the wave propagation
4	Amplitude _____	D	Number of waves per second
5	Trough E.g G	E	Distance from crest to next crest
6	Frequency _____	F	Particle oscillation is parallel to the wave propagation
7	Period _____	G	Lowest position of oscillating particles
8	Wave length _____	H	Time for one wave
9	Speed _____	I	$V = \lambda F$
10	Wave equation _____	J	Distance per unit time

6. A load of 20 N is hung from a spring that obeys hook's law.



A. Write down hook's law .

(2 Marks)

.....
.....

B. If the spring stretches by 0.04m , calculate the spring constant? (3 Marks)

.....
.....
.....

- C. The property of a spring to return its original position after deforming forces are removed is known as (Elasticity, ductility) . (2 Marks)

.....
.....

- D. Calculate the extension if a force of 50 N was applied ? (3 Marks)

.....
.....
.....
.....

7. A. Define the term : " half life" of radioactive sample

(2 Marks)

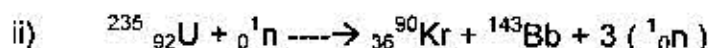
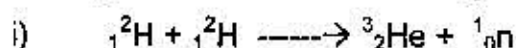
.....
.....

- B. In an experiment to find the half life of radioactive sample , the count rate falls from 400 Bq to 50 Bq in 15 days. Work out the half life of the sample.

(3 marks)

.....
.....
.....

- C. Distinguish between the following nuclear reactions :



(2marks)

Ans: Reaction 1 stands for :

Reactions 2 stands for :

- D. State any three positive uses of radioactivity?

(3 Marks)

1.
2.
3.

8. A) Define Momentum and give its unit

(2 Marks)

.....
.....
.....

B. The equation $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ is mathematical expression for the law of conservation of linear momentum. Describe the law in words. (2 Marks)

.....
.....
.....

C. A body of mass 10 Kg is moving at 4 m/s . Calculate its momentum ? (3Marks)

.....
.....
.....

D.What will happen to the momentum if the body becomes at rest. Tick one box:

Increases	☐
Decreases	☐
Becomes Zero	☐
Does not change	☐

(2 Marks)

Somalilandschools.Online

**FREE EXAM PAPERS,
LESSONS, NOTES AND**

TEXT BOOKS

Contact:

00 252 63 6551739

00 252 63 4421900

00 252 65 4421900

END